

Polynomials



1

- a Yes, it is a polynomial.
- c No, it is not a polynomial.
- e No, it is not a polynomial.
- g No, it is not a polynomial.

- b Yes, it is a polynomial.
- d Yes, it is a polynomial.
- f No, it is not a polynomial.
- h Yes, it is a polynomial.

2

- a
- i 7
 - ii 2
 - iii 2

- b
- i 5
 - ii $-\sqrt{5}$
 - iii $7\sqrt{6} + 5$

- c
- i 7
 - ii $\frac{1}{5}$
 - iii 5

- d
- i 3
 - ii $-\frac{7}{10}$
 - iii 6

Add and subtract polynomials

3

- a $2x^3 - 6x^2 - x + 4$
- c 2

- b 3
- d Yes

4 $P(x) + Q(x) = 5x^2 - 8x + 2$

5 $P(x) - Q(x) = 3x^2 + x + 1$

6 $B(x) - A(x) = 6x^2 + 4x - 1$

7 $A(x) + B(x) + C(x) = 4x^2 - 4x$

8 $A(x) - B(x) - C(x) = 7x^2 - 4x - 1$

9

a $4x^3 - 4x^2 - 10x - 4$

b $-5x^3 - 4x^2 - 10x - 1$

10



a $-12x^2 - 8x - 3$

c $\frac{5}{6}x^2 + \frac{3}{20}x + \frac{1}{4}$

e $-7x^2 - 7x + 26$

g $2x^2 - x - 13$

i $-2x^3 - 3x^2 + 9x + 5$

b $-4x^3 - 9x^2 - 17x - 7$

d $-2x^3 + 6x^2 + 7$

f $-9x^3 + 4x^2 + 3x - 1$

h $-11x^3 + 9x - 2$

j $3x^3 - 8x - 7$

11 $a = 1, b = 12$

Multiply polynomials

12

a $x^2 + 7x + 10$

c $ab - 6a + 9b - 54$

e $5m^2 - 7m - 24$

g $35w^2 + 39w + 10$

i $-90y^2 + 26y + 48$

k $-x^2 - 7x - 10$

m $-4y^2 + 12y + 112$

b $x^2 - 5x + 6$

d $m^2 - 3m - 18$

f $42y^2 + 6y - 36$

h $\frac{1}{9}y^2 + \frac{1}{36}y - \frac{1}{24}$

j $9y^2 + 90y + 144$

l $-5y^2 - 45y - 100$

n $-75x^2 + 110x - 40$

13

a $-30x^2 - 42x + 34$

c $5a^3 + 8a^2 - 2a + 4$

e $-25a^3 - 50a^2 - 45a - 20$

g $-12m^3 - 20m^2 + 8$

b $10n^2 + 29n + 6$

d $u^3 + u^2 - 15u - 20$

f $20c^4 - 15c^3 + 2c^2 + 6c - 4$

h $-56x^2 + 24xy - 12x + 36y + 108$

14 True

15 The linear x terms will come from multiplying constants with terms in x . In particular, the x term will be:

$$(-3x)(-4) + (5x)(-5) = -13x$$

So, the coefficient of x will be -13 .

16

a 12

b -18



c Yes

17 $(x + 6)(x + 2)$

Applications

18

a $8x^2 + 18x + 14$ units

b $4x^3 + 6x^2 + 3x + 8$ units

c $16x^2 + 20x + 18$ units

d $(6x^2 + 14x + 24)$ units

e $(3x^2 + 24x)$ units

f $(3x^3 + 8x^2 + 4x + 9)$ units

19 $12x^3 + 34x^2 - 18x + 6$

20 $42x^2 + 59xy + 20y^2$

21 $8y^3 + 22y^2 + 17y + 3$ ft³

22 $12x^2 + 144x + 384$ m³

23 $60x - 32x^2 + 4x^3$ in³

24

a $h + 9$ in

b $w + 8$ in

c $4h + \frac{9w}{2} + 36$ in

25 $(128y^2 + 48y)$ ft³